Worksheet: Mixtures and Solutions

Name: _____ Class: _____

1. State, with a reason, whether each of the following is a homogeneous mixture (solution) or a heterogeneous mixture (mechanical mixture):

a. hotdog relish

b. freshly squeezed grapefruit juice

c. pop

d. apple juice

e. granola

f. vegetable soup

2. Give an example of the following:

a pure substance

b. a solid solution

c. a liquid solution

d. a gas solution

e. a mechanical mixture

3. Susan makes the following entry in her notebook: “On Friday we were given a blue liquid in a shallow container. We placed it on the windowsill over the weekend. On Monday morning, there was no liquid left, but the dish had some solid blue stuff in it.”

a. Was the blue liquid in the dish a heterogeneous mixture, a solution, or a pure substance? Explain your choice.

b. Write a few sentences to explain what happened in the dish.

4. Use the particle theory to explain:

a. The difference between a pure substance and a mixture.

b. The difference between a solution and a mechanical mixture.

5. Describe whether each of the following is a solution or not. Give reasons for your choice:

a. a mixture of clay and water

b. a mixture of salt and water

c. tomato juice

6. The Moon was once believed to have no water anywhere on its surface. If this were true, could there be solutions on the Moon? Explain.

7. Draw diagrams to show the differences between:

a. Pure substances and mixtures

b. heterogeneous and homogeneous mixtures

8. Match the Terms to their definitions.

Terms	**Definitions**
A. Matter B. Properties C. Mixture D. Heterogeneous E. Homogeneous F. Mechanical mixture G. Homogeneous mixture (solution) H. Pure substance I. Dissolving J. Evaporation K. Solute L. Solvent M. Soluble N. Insoluble	_____ made up of visible parts _____ a mixture that appears the same all over and the individual materials are not visible _____ the change from a liquid state to a gas state _____ <u>not</u> able to be dissolved in a particular solvent _____ any material that occupies space and has mass _____ a material that has more than one set of properties _____ any material made up of only one kind of extremely small particle _____ the characteristics used to describe matter _____ forming a solution by mixing two or more materials _____ every part of the material is the same (only one set of properties) _____ the substance that dissolves in a solvent to form a solution _____ able to be dissolved in a particular solvent _____ the substance that dissolves a solute to form a solution _____ a mixture where the different parts can be identified easily with the unaided eye